FIT5230 Assignment & Milestones (50%)

Malicious AI

Assignment Description

For this assignment, you will work in **teams of 2**, or you may choose to complete it individually. To ensure that every team has the chance to both attack and defend their designed frameworks, each theme will be capped at **20 teams, split evenly between 10 Light (Defense) teams and 10 Dark (Attack) teams**. This balance is essential to maintain fairness and engagement across all themes, so teams are encouraged to organize early to secure their preferred theme and side. The assignment is divided into **4 milestones**, each designed to guide you through the process step by step, from initial planning and framework design to execution, attack/defense preparation, and final presentation. Detailed instructions, deliverables, and deadlines for each milestone will be provided to help you stay on track and successfully meet the assignment objectives.

How to start

1. Choose a **theme** and select a **side** (Light vs Dark):
    - **Theme 1: Adversarial ML on Gen AI**:
      Adversarial machine learning (ML) against generative AI is generally regarded as a malicious activity by default, as it aims to attack, manipulate, or compromise the behaviour of generative AI systems. The roles are defined as follows:
        - **Light**: How to defend against adversarial ML attacks on generative AI?
            - If your selected paper focuses on a defence or mitigation technique against adversarial machine learning attacks on generative AI, you should improve upon the existing technique to enhance its effectiveness, robustness, or practicality.
            - If your selected paper focuses on an adversarial machine learning technique targeting generative AI, you should propose a method to counter or defend against the attack presented in the paper.
        - **Dark**: How can existing adversarial ML attacks be improved to become more effective and harder to detect?
            - If your selected paper focuses on an adversarial machine learning technique targeting generative AI, you should improve upon the proposed adversarial technique to make it more effective or robust.
            - If your selected paper focuses on a defence or mitigation technique against adversarial machine learning attacks on generative AI, you should develop an adversarial attack strategy to evaluate and potentially bypass the proposed defence mechanism.
        - Sample paper(s):
            - WMCodec (sample code)
            - AdvSticker (sample code)

- **Theme 2: Text-to-Image (TTI)**:
  Text-to-image (TTI) models are considered a dual-use generative AI technology, as they provide legitimate creative applications while also posing risks when exploited by malicious actors to generate realistic fake images. The roles are defined as follows:
  - **Light**: The defender is an entity responsible for improving the security, robustness, and trustworthiness of TTI models by detecting, mitigating, or preventing adversarial attacks.
    - If your selected paper focuses on enhancing the security, robustness, or trustworthiness of TTI models, your task is to further improve the proposed technique to make it more effective, robust, or practical.
    - If your selected paper focuses on attacking or exploiting TTI models, your task is to develop a method to defend against or mitigate the attack technique presented in the paper.
  - **Dark**: The attacker is an adversarial entity that seeks to exploit vulnerabilities in TTI models to manipulate, deceive, or compromise the generated outputs or the model itself.
    - If your selected paper focuses on attacking or exploiting TTI models, you should improve the proposed attack technique to make it more effective, robust, or practical.
    - If your selected paper focuses on enhancing the security, robustness, or trustworthiness of TTI models, you should design an attack technique to evaluate and potentially bypass the defence mechanism proposed in the selected paper.
  - Sample paper(s):
    - Safe Latent Diffusion (sample code)
    - InstructPix2Pix (sample code)

- **Theme 3: Speech-to-Face (S2F)**:
  Speech-to-Face (S2F) models represent a dual-use generative AI technology that enables legitimate applications while introducing potential security and privacy risks when misused by malicious actors to generate realistic synthetic media. The roles are defined as follows:
  - **Light**: The defender is responsible for improving the security, robustness, and trustworthiness of S2F models by developing techniques to detect, prevent, or mitigate potential misuse and attacks.
    - If your selected paper focuses on enhancing the security, robustness, and trustworthiness of S2F models, you should propose further improvements to the technique presented in the selected paper.
    - If your selected paper focuses on exploiting vulnerabilities in S2F models, you should develop a technique to mitigate or defend against the attack described in the selected paper.

- **Dark**: The attacker attempts to exploit vulnerabilities in S2F models to generate, manipulate, or misuse synthetic facial representations derived from speech inputs.
  - If your selected paper focuses on exploiting vulnerabilities in S2F models, you should propose further improvements to the attack technique presented in the selected paper.
  - If your selected paper focuses on enhancing the security, robustness, and trustworthiness of S2F models, you should propose an attack method to evaluate and potentially bypass the defence technique described in the selected paper.
- Sample paper(s):
  - GenVidBench (sample code)
  - SadTalker (sample code)
  - Hallo (sample code)

Note: *You are free to choose other papers and codes apart from the provided samples as long as they belong to the chosen theme*

2. Choose a team **name**: Light.teamname *e.g. Light.Soothsayers*

3. **Familiarize** yourselves with the selected Colabs/GitHubs
   - Discuss among yourselves (or with yourself if you're doing this individually) how you want to **vary** the techniques in one of those reference codes

Key Action*:* Please email your tutor with the following details:

1. Team name
2. Team member names
3. Selected theme
4. Chosen side (Light or Dark)

Note*:* *Confirmed teams will be listed in the spreadsheet [here]. Please check the list before emailing to ensure there are available slots for your preferred theme and side. Teams that do not inform the tutor and are not listed on the spreadsheet by the end of Week 4 will be randomly assigned to an available theme and/or side.*

1st Milestone (2%): Throwing Down the Gauntlet

Milestone 1 due: 28 August 2026, 11:55PM MYT (Week 5)

1. This milestone is a **group-based** assessment task.
2. Each team to **post** to the FIT5230 Ed Discussion forum, the following:
    - Forum heading: "[teamname] topic"
    - Your **team's name** and team members' **names** and **photos** of members
    - Provide the **reference** material (i.e., Colab/GitHub link) and the main research paper your work is based on to serve as a baseline.
    - Explain why the chosen problem is interesting or challenging.
    - Design and describe an interactive challenge for other teams to attempt. This could involve:
        - Testing Robustness: e.g., "Can you break our defense model?" by bypassing your security measures or finding adversarial inputs.
        - Detection Tasks: e.g., "Can you detect our attacks?" by identifying crafted malicious inputs or hidden patterns in your data.
    - Share your team's Colab link (it doesn't need to significantly differ from the reference Colab/GitHub, but it must be your own version).
3. What to submit:
    i. **Ed Post** on Ed Forum
    ii. **Link of Ed Post** included in a PDF or text file to Moodle Milestone 1 submission page
4. **Criteria for marks:**

Criteria	Marks	Description
Baseline Justification	0.5%	Is there a strong rationale for selecting the reference model or paper, and does it explain why the challenge is interesting or meaningful?
Clarity of Problem Statement	0.5%	Is the problem well-defined and technically relevant?
Challenge Design Quality	0.5%	Is the interactive task for other teams thoughtfully crafted, original, and non-trivial?
Initial Customization & Setup	0.4%	Is there evidence of modification beyond just cloning the reference code?
Clarity of the Ed Post	0.1%	Is the Ed post well-written, structured, and easy to follow?

2nd Milestone (8%): Show of Force

Milestone 2 due: 18 September 2026, 11:55PM MYT (Week 8)

1. This milestone is a **group-based** assessment task.
2. **Posts** to the FIT5230 Ed Discussion forum, on the following:
    - Main results so far: **changes** you've made to the reference Colab incl some **demos**
    - A **functional base code** that enables other teams to perform attacks or defenses.
    - Which aspect of the other team are you **targeting**, e.g., their ideas or any demonstrations.
3. What to submit:
    i. **Ed Post** on Ed Forum
    ii. **Link of Ed Post** included in a PDF or text file to Moodle Milestone 2 submission page
4. **Criteria for marks:**

Criteria	Marks	Description
Quality of description	2%	Is the problem clearly stated? Are the methods, challenges, and solutions explained logically and concisely?
Technical depth	3%	Are the models, algorithms, or methods applied in the functional base code beyond basic implementation?
Documentation and Reproducibility	1%	Marks for well-organized documentation, clear instructions, and making the project easy for others to run.
Engagement with other teams	2%	Did the team have meaningful attacks/defenses? Did they respond to others' ideas thoughtfully?

3rd Milestone (25%): Champions of the World

Milestone 3 due: 22 October 2026, 11:55PM MYT (Week 12)

1. This milestone is a **group-based project task** with **individual assessment components**.
2. Give a **15-minute presentation** during the workshop/tutorial session covering the following topics, with all team members taking turns to present:
    - Present the **key outcomes** achieved to date and describe the **progress** and **improvements** made to your proposed method since Milestone 2. (Approximately 3 minutes)
    - Present **ideas** you've tested on other teams' targeted aspects, or experiments based on the Colab/techniques they referenced. (Approximately 2 minutes)
    - Each team member should present **one** distinct strategy used to defend against or attack other teams' work (e.g., blindsiding, deception, or keen observation). Explain how the team collaborated to develop and apply the mentioned strategy, or, for individual projects, how you implemented the idea in your work. Each member should take a turn presenting, and the entire presentation for this section should be completed within 10 minutes.
3. What to submit:
    i. **Link of Google Colab** included in a PDF or text file to Moodle Milestone 3 submission page
4. **Criteria for 15-minute presentation (20%):**

Criteria	Marks (%)	Description
Framework Progress & Outcomes	5	Present the key outcomes achieved since Milestone 2. Explain the technical rationale behind design choices and modifications.
Active Peer Engagement	3	Demonstrate active engagement with ideas, code, or Colabs shared by other teams.
Individual Strategy: Design & Technical Depth	4	**Evaluated individually per student**. Articulate a distinct strategy used to attack or defend against another team's work (e.g., exploitation of logical flaws, deception, or blindsiding).
Individual Strategy: Critical Analysis & Impact	5	**Evaluated individually per student.** Critically evaluate the outcome and real-world impact of the strategy. Analyze why it succeeded or failed, how the target team reacted/countered, and what key technical lessons were learned from the interaction.
Individual Strategy: Integration & Role	2	**Evaluated individually per student.** Clearly explain your specific personal contribution to developing, testing, and deploying this

		strategy, and how you coordinated with teammates to integrate it into your team's overall execution.
Presentation quality	1	Evaluate professional delivery, clarity, slide organization, time management (fitting within the 15-minute allocation), and responsiveness during live engagement.

5. **Criteria for Colab submission (5%):**

Criteria	Marks (%)	Description
Execution & Output Completeness	2	The notebook demonstrates a complete, functional workflow from start to finish. All relevant code cells must display pre-rendered execution outputs (e.g., logs, metrics, visualizations) that substantiate the findings and demonstrate that the workflow ran successfully end-to-end.
Technical & Logical Rigor	1	Implementation of data handling, algorithms, pipelines, and evaluation procedures is technically sound. Demonstrates appropriate methodology while avoiding common analytical or coding errors (such as data leakage, incorrect validation setups, or broken logic).
Reproducibility & Execution Readiness	1	Dependencies, environment configuration, and external resources are managed so that the code can run seamlessly end-to-end without broken package requirements, missing file references, or environment-specific hardcoded paths.
Notebook Narrative & Documentation	1	The notebook is structured as a self-contained report. Markdown cells and comments clearly articulate the problem being solved, the methodology used, the key results obtained, and potential future directions alongside the corresponding code cells.

4th Milestone (15%): Adversarial Gameplay

Milestone 4 due: 2 November 2026, 11:55PM MYT (Week 14)

1. This milestone is an **individual-based** assessment task.
2. Each student to submit an individual report written with the Overleaf [www.overleaf.com] using the IEEE Transactions journal template:

   https://www.overleaf.com/latex/templates/ieee-latex-template-for-transactions-on-magnetics/hncvmwqcydfn

3. In the report, each student (not team) should keep a weekly log between Week 2 to Week 12, concisely describing different strategies the student has used throughout the semester in order to maximize his/her marks & advantage over other students & teams including his/her own teammates.
4. After the weekly log, the report should also consist of the following unique forms of adversarial gameplay in the last section of the report:
   - Discuss what you would do differently to **enhance** the system if there were no time or resource constraints. Include a proof of concept or demo to show that your idea is feasible.
   - Share your **personal reflection**, including your background, and explain how it influenced your strategies or provided an advantage in tackling the selected theme and coding tasks in this assignment.
   - If you were to switch to the **opposite side** of your current Google Colab project, outline the strategies you could use to attack (if you initially defended) or defend (if you initially attacked). Provide a proof of concept or demo to demonstrate the concept.
5. Each student **submits** the report to the Milestone 4 Moodle submission page.
6. **Criteria for marks:**

Criteria	Marks (%)	Description
Weekly Strategy Log (Weeks 2–12)	3	Clear, concise weekly record documenting individual strategies used to maximize performance and maintain a competitive advantage over other teams/teammates. Shows continuous reflection, strategic adaptation, and individual engagement across the semester.
Unconstrained System Enhancement & Feasibility	4	Describe a specific idea for a new feature, architecture, or mechanism. Provide a functional demo (e.g., code snippet, Colab notebook, video) to show the idea's feasibility. Explain the technical details and why it's a significant improvement.

Adversarial Role-Reversal Strategy & Demo	5	Outline specific strategies for attacking (if you were a defender) or defending (if you were an attacker). Provide a functional demo (e.g., code snippet, Colab notebook, video) to showcase a key part of your new strategy. Justify why this strategy would be effective, based on your knowledge from the original project.
Personal Reflection & Background Alignment	1.5	Authentic reflection connecting the student's personal/technical background to their chosen strategies and project execution. Clear self-awareness regarding skills developed, lessons learned, and unique advantages leveraged during the task.
Analytical Rigor	1.5	The report strictly follows the required IEEE Transactions Overleaf template. All claims, vulnerabilities, and strategic decisions are backed by data, rigorous logic, and deep critical analysis rather than superficial observations.